IMPERIAL

CCIMA Centre for Climate Impact Adaptation

BCKIC Bhubaneswar City Knowledge Innovation Cluster Foundation

Unlocking the Potential *of*

Combining Carbon Markets with Biodiversity

Problem Statement

- ***To date, it has been difficult to attach a quantitative financial value to biodiversity***, in spite of its qualitative value to various critical ecosystem services. Conversely, carbon sequestration is relatively easy to quantify and financially value. As a result, carbon markets have taken off, while biodiversity is mostly treated as an optional co-benefit, *even though restoring biodiversity is vital for meaningful climate adaptation and resilience*. Thus, ***how does one attract financing for essential biodiversity restoration?***

- Nature-based Solutions (NbSs), whether for carbon storage or other ecosystem services, ***mostly focus on adapting to the present climate***. However, the point of climate change is that the climate is *changing*, at an accelerating pace. Thus, ***how does one optimise adaptation solutions for future climate?***

Solution: GEMS (Global Ecosystem Model Services)

- **Scientific Basis**: Recent ground-breaking study of the *link between biodiversity and ecosystem services*, carried out at **Imperial College London**. The analysis (of > 300,000 data points, covering ecosystems in all seven continents and all ecosystem services in the IPBES list; Fig. 1a) **showed that**:

***Biodiversity quantifiably controls the amount of carbon sequestration in any ecosystem*, as well as many other essential ecosystem services** (Figs. 1b,c). Hence:

(1) Biodiversity can be **directly valued financially** by how much carbon sequestration it enables; and

(2) The future effect of climate change on carbon sequestration (and other ecosystem services) can be **directly predicted** by the effect of climate change on biodiversity.

GEMS operationalises these findings, based on the data, analyses and results of this study.

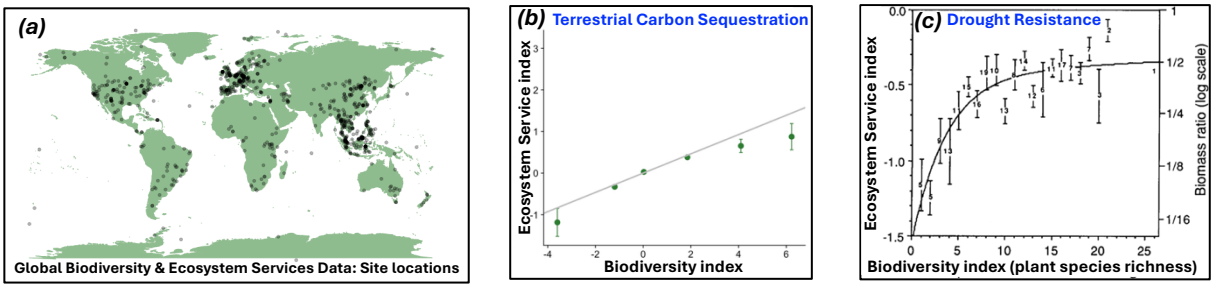


Fig. 1: *(a)* Global site locations of study's biodiversity & ecosystem services data. *(b)* & *(c)*: Dependence of two critical ecosystem services (terrestrial carbon sequestration and drought resistance, respectively) on biodiversity.

Study link: *Biodiversity safeguards ecosystem services and functions worldwide*: Moffet et al., **Nature** Eco & Evo, **2026**

What Outcomes Does GEMS Enable?

GEMS ***quantitatively predicts the amount of carbon storage*** (and other pertinent services) for any given ecosystem, ***based on the biodiversity*** (current or future) of the ecosystem.

As such, GEMS enables the following **specific outcomes:**

- **Current C-storage**: Provides a *quantitative estimate of the current amount of carbon sequestration* in any ecosystem, based on its current biodiversity content.

- **Scenario-based Future C-storage**: The existing PREDICTS model estimates future biodiversity content in ecosystems, under various SSP and RCP socio-economic and climate projections. GEMS incorporates these biodiversity projections to provide a *quantitative estimate of the future carbon storage in any ecosystem under different socio-economic and climate scenarios*.

- **C-storage Potential**: Provides a *quantitative estimate of the maximum carbon-sequestration possible* in any ecosystem, under ideal conditions.

- **Other Ecosystem Services**: *Pinpoints which services* (besides carbon storage) *are maximally affected by biodiversity in any given ecosystem,* and *provides quantitative estimates of these services* under current, future or ideal conditions.

- **MRV**: *Provides an integrated methodology, based on biodiversity indices, to Monitor, Report and Validate changes in a range of ecosystem services* (including carbon storage) in any given ecosystem.

Combining the above, GEMS enables the following **critical overarching outcomes:**

(1) ***Financing of biodiversity restoration via carbon markets***, by directly and *quantitatively* linking changes in biodiversity to changes in the amount of carbon sequestered in any ecosystem.

(2) ***Optimisation of Nature-based Solutions for a desired suite of services in any ecosystem*** (natural, man-made, or hybrid), by identifying how the individual ecosystem services vary with biodiversity.

(3) ***Optimisation of Nature-based Solutions for future climate***, by identifying how the targeted ecosystem services vary with biodiversity under different climate change and socio-economic scenarios.

(4) ***Simplification of MRV protocols,*** by providing a single unified MRV methodology, based on biodiversity indices, for diverse ecosystem services.

In summary, GEMS offers an unprecedented means of combining biodiversity restoration, carbon markets and climate adaptation, that optimises both financial viability and long-term climate resilience.